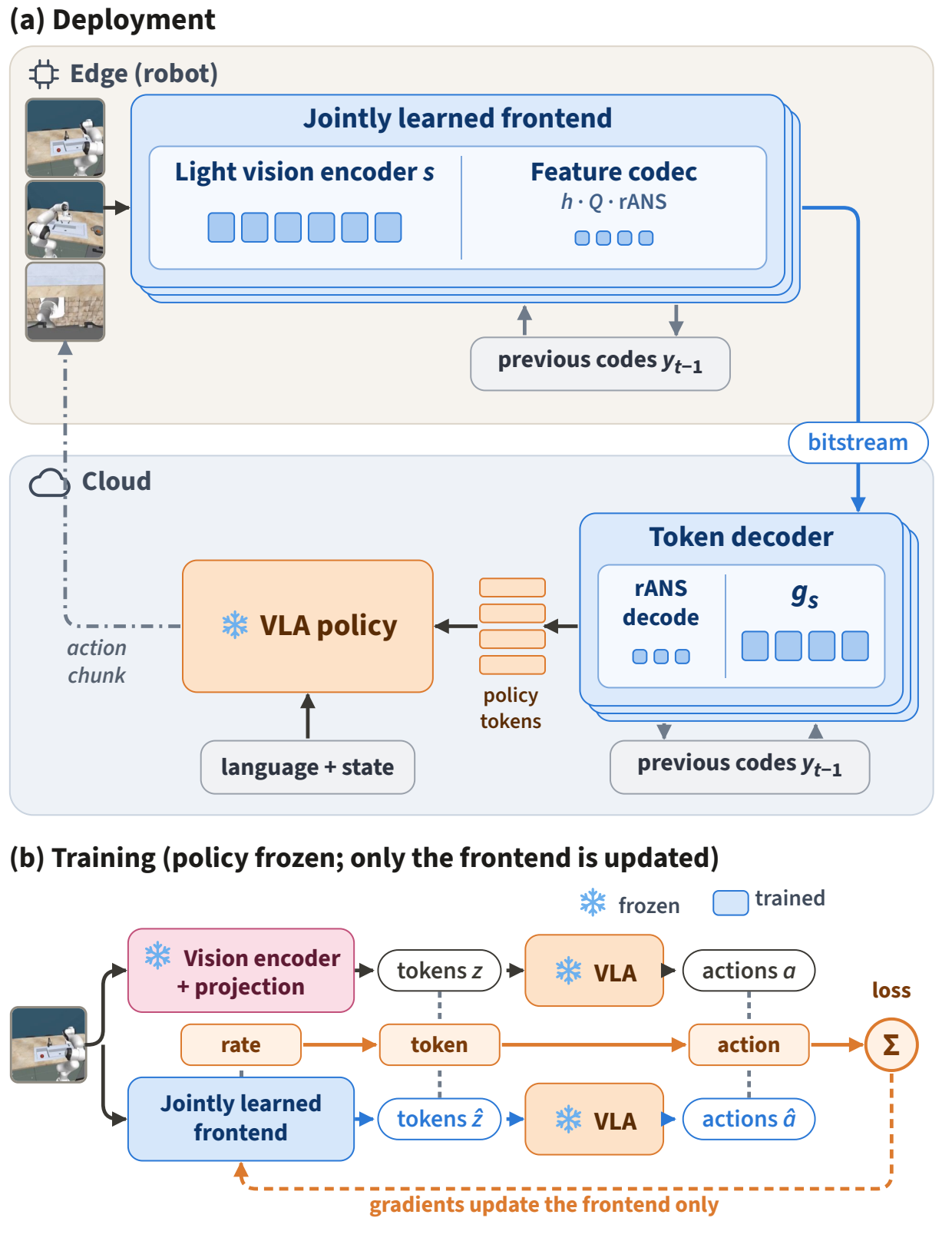

(a) Deployment
Edge (robot)
Jointly learned frontend
Light vision encoder s
Feature codec
h · Q · rANS
previous codes y_{t-1}
bitstream
Cloud
Token decoder
rANS decode
g_s
VLA policy
action chunk
policy tokens
language + state
previous codes y_{t-1}
(b) Training (policy frozen; only the frontend is updated)
frozen
trained
Vision encoder + projection
tokens z
VLA
actions a
loss
rate
token
action
Σ
Jointly learned frontend
tokens ẑ
VLA
actions â
gradients update the frontend only